

Predicting Myocardial Infarction from Cardiac Features: A Multi-Modal Ablation Study

Jim Titus

June 10, 2026

Contents

1	Overview	1
2	Methods and analysis	2
2.1	Data ingestion	2
2.2	Data cleaning and target preparation	3
2.3	Feature bucketing: the core design choice	3
2.4	Exploratory data analysis	4
2.5	Train/test split	7
2.6	Modeling approach	8
2.7	The ablation	9
3	Results	10
4	Conclusion	11
5	References	11

1 Overview

This project began with the HEART score, a clinical tool that stratifies a chest-pain patient's risk of a major cardiac event from five inputs (History, ECG findings, Age, Risk factors, and Troponin). Historically, emergency departments struggled with intake decisions for patients presenting with chest pain. Bed and clinician capacity are fixed constraints, yet sending these patients home risks denying care for acute cardiac incidents. The HEART score's grouping of predictors into a small set of clinically meaningful buckets is what shaped both the dataset choice and the modeling approach for this study.

This dataset's feature coverage aligns well with the HEART inputs. It carries rich ECG findings and age alongside the standard cardiac risk factors, even though it lacks a troponin biomarker and a structured patient history. The coverage is good but deliberately imperfect, which makes the question of what is predictable without troponin a real one. It also has little prior exposure on Kaggle, which fits the project requirement to avoid well-known or heavily reused data. Perhaps most importantly, it has 5,000 patient records across

roughly 40 candidate predictors. That's more rows and features than many comparable cardiac datasets, which gives the models ample data to train and test on.

The outcome is a binary `mi_label`, indicating whether the patient had a myocardial infarction (MI, a heart attack). It is present in about 18.5% of patients. The predictors fall into three natural HEART-style groups: standard demographic and clinical risk factors, electrocardiogram (ECG) waveform features, and photoplethysmogram (PPG) features (the kind a wrist or fingertip wearable can capture). A fourth bucket of measurement-quality metadata was excluded, as it describes the recording rather than the patient.

A first model using the HEART-aligned features classified MI almost perfectly, around 99.9% accuracy. That accuracy comes from the data more than the model. The ECG features encode the diagnostic markers of MI directly. ST-segment elevation, T-wave changes, and pathological Q-waves are the classic criteria a cardiologist uses, and a synthetic data generator encodes them more cleanly than real physiology ever would. A clean ECG makes MI detection trivial here.

That near-perfect accuracy is what prompted the actual investigation. Rather than report an uninteresting 99.9%, I re-bucketed the predictors and began removing features to see how much accuracy I could retain without the diagnostic ECG channel. This is an ablation study. So the core question quickly became how well the non-diagnostic, wearable-grade PPG features + basic demographics could predict MI on their own.

The key steps performed were to clean and structure the data, assign every predictor to a clinical bucket (demographics, ECG, PPG, or measurement-quality metadata), explore the data to locate where the class separation lives, define a small family of models by which buckets each is allowed to see, and train two algorithms (logistic regression and a random forest) on each feature set. Model comparison leans on metrics that stay honest under class imbalance rather than on accuracy alone.

2 Methods and analysis

2.1 Data ingestion

The data (see references for Kaggle citation) can be downloaded automatically from my public GitHub repository, so the analysis is fully reproducible with no manual download or login. The raw file has 50 columns and 5,000 rows.

```
# Auto-install any missing packages, then load.
if (!require(caret, quietly = TRUE))
  { install.packages("caret"); library(caret) }
if (!require(randomForest, quietly = TRUE))
  { install.packages("randomForest"); library(randomForest) }
if (!require(ggplot2, quietly = TRUE))
  { install.packages("ggplot2"); library(ggplot2) }

# Download once (relative path); reuse the local copy on re-runs.
data_url  <- paste0("https://raw.githubusercontent.com/leftbrain847/",
  "Myocardial_Infarction_Prediction/main/Data/combined_cardiac_dataset.csv")
local_copy <- "combined_cardiac_dataset.csv"
if (!file.exists(local_copy)) download.file(data_url, local_copy, mode = "wb")
cardiac <- read.csv(local_copy, stringsAsFactors = FALSE)

dim(cardiac)
```

```
## [1] 5000  50
```

2.2 Data cleaning and target preparation

Two columns are removed before modeling. `patient_id` is a pure identifier with no predictive content, and `mi_subtype` is an alternate, more granular version of the target (it labels MI cases as STEMI, NSTEMI, or Old_MI). Leaving `mi_subtype` in would leak the outcome directly, so I dropped it. The binary target `mi_label` is converted to a labeled factor ("No"/"Yes") so that `caret`'s classification metrics report sensitivity and specificity against the clinically meaningful positive class. Remaining text columns (for example `sex` and `smoking_status`) are converted to factors. The dataset has no missing values, so no imputation is required.

```
# Drop the identifier and the leak-prone alternate target.
df <- cardiac[, !(names(cardiac) %in% c("patient_id", "mi_subtype"))]

# Target -> factor with "Yes" as the positive class.
df$mi_label <- factor(ifelse(df$mi_label == 1, "Yes", "No"), levels = c("No", "Yes"))

# Encode character categoricals as factors.
chr_cols <- names(df)[sapply(df, is.character)]
df[chr_cols] <- lapply(df[chr_cols], factor)

# Confirm there is nothing to impute.
cat("Total missing values:", sum(is.na(df)), "\n")
```

```
## Total missing values: 0
```

```
cat("MI prevalence:", round(mean(df$mi_label == "Yes"), 4), "\n")
```

```
## MI prevalence: 0.1848
```

2.3 Feature bucketing: the core design choice

Every predictor is assigned to exactly one of four clinical buckets. Models are then defined by choosing which buckets they are allowed to look at. This makes the ablation transparent and easy to extend.

The buckets are:

- **Demographics** age, sex, BMI, smoking status, diabetes, hypertension, cholesterol, family history.
- **ECG** the electrocardiogram morphology features, including the diagnostic ST/T/Q-wave measurements, plus heart-rate-variability (HRV) metrics, which are ECG-derived.
- **PPG** the photoplethysmogram features a wearable optical sensor produces (oxygen saturation, pulse transit time, pulse-wave velocity, perfusion, waveform shape).
- **Signal quality** measurement-quality metadata (noise level, recording lead, motion artifact). This describes the recording rather than the patient, so letting it predict MI would be an artifact. It is excluded from every model.

Two bucketing decisions are worth flagging because they make the central comparison harder rather than easier, and therefore more honest. First, the HRV metrics are placed in the ECG bucket because they are computed from the ECG R-R intervals. Second, `pulse_rate_bpm` and `peak_to_peak_interval_ms` are measured by the PPG sensor but carry the same cardiac-cycle (heart-rate) information as the ECG, with which they are highly collinear. I assigned them to ECG. This deliberately handicaps the PPG-only model by denying it heart-rate access, so any performance it retains comes from genuinely PPG-specific information rather than a back-door copy of heart rate.

```

buckets <- list(
  demographics = c("age", "sex", "bmi", "smoking_status", "diabetes",
    "hypertension", "total_cholesterol_mg_dl", "family_history_cad"),

  ecg = c("heart_rate_bpm", "rr_interval_ms", "pr_interval_ms", "qrs_duration_ms",
    "qt_interval_ms", "qtc_bazett_ms", "st_elevation_mv", "st_depression_mv",
    "st_slope", "t_wave_amplitude_mv", "t_wave_inversion", "pathological_q_wave",
    "p_wave_duration_ms", "p_wave_amplitude_mv",
    # HRV is ECG-derived:
    "hrv_sdnm_ms", "hrv_rmssd_ms", "hrv_lf_power_ms2", "hrv_hf_power_ms2",
    "hrv_lf_hf_ratio"),

  ppg = c("spo2_percent", "pulse_transit_time_ms", "augmentation_index_pct",
    "stiffness_index_m_s", "reflection_index_pct", "systolic_peak_amplitude",
    "diastolic_peak_amplitude", "pulse_wave_velocity_m_s", "perfusion_index_pct",
    "systolic_upstroke_time_ms", "diastolic_time_ms", "crest_time_ratio"),

  signal_quality = c("signal_quality_score", "noise_level_db", "recording_lead",
    "ppg_signal_quality", "motion_artifact_score", "measurement_site")
)

# pulse_rate_bpm and peak_to_peak_interval_ms are PPG-measured but heart-rate-
# equivalent and collinear with ECG heart rate, so they go to ECG (a deliberate
# handicap on the PPG-only model, explained above).
buckets$ecg <- c(buckets$ecg, "pulse_rate_bpm", "peak_to_peak_interval_ms")

# Integrity check. Every predictor must land in exactly one bucket, none missing
# and none duplicated. If this fails the analysis stops rather than silently
# dropping or double-counting a feature.
all_bucketed <- unlist(buckets, use.names = FALSE)
all_preds <- setdiff(names(df), "mi_label")
stopifnot(setequal(all_bucketed, all_preds))
stopifnot(!any(duplicated(all_bucketed)))

sapply(buckets, length)

```

```

##    demographics          ecg          ppg signal_quality
##              8             21          12              6

```

2.4 Exploratory data analysis

The first thing to establish is the class balance, because it dictates which metrics are trustworthy later. MI is present in about 18.5% of patients, so a model that does nothing but predict “No” for everyone would already be about 81.5% accurate. Accuracy alone is therefore close to meaningless here. The analysis must lean on sensitivity (the share of true MIs caught), specificity, balanced accuracy, and area under the ROC curve (AUC).

```

ggplot(df, aes(mi_label, fill = mi_label)) +
  geom_bar(colour = "black") +
  scale_fill_manual(values = c("No" = "lightsteelblue", "Yes" = "indianred")) +
  labs(title = "Class balance of MI label", x = "MI", y = "Patients") +
  theme_classic() +
  theme(legend.position = "none")

```

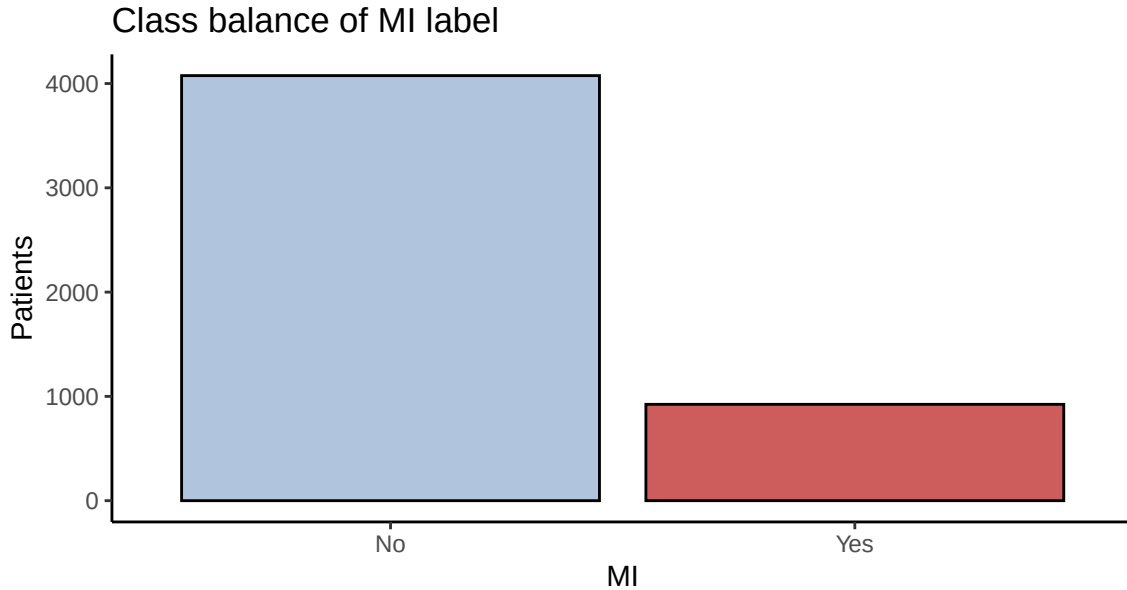


Figure 1: Class balance. MI is the minority class at about 18.5 percent, so accuracy must be read against a no-information rate of roughly 0.815.

The ECG diagnostic features separate MI from non-MI almost completely, and ST-segment elevation shows it most clearly. The two distributions barely overlap. That's not a leak or a modeling artifact. The synthetic generator built the textbook ECG criteria for MI straight into these columns, so the clean split is a property of the data itself. Seeing this plot makes the later 99.9% accuracy unsurprising rather than suspicious.

```
ggplot(df, aes(st_elevation_mv, fill = mi_label)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("No" = "lightsteelblue", "Yes" = "indianred")) +
  labs(title = "ST elevation by MI label (ECG feature)",
       x = "ST elevation (mV)", y = "Density", fill = "MI") +
  theme_classic()
```

The PPG features carry real but weaker separation. Pulse-wave velocity, a marker of arterial stiffness an optical sensor can estimate, shifts clearly higher in MI patients. The overlap is larger than for ST elevation, so PPG alone should predict well but not perfectly. This previews the central finding, that the wearable channel recovers a great deal of the predictive power even without the diagnostic ECG.

```
ggplot(df, aes(pulse_wave_velocity_m_s, fill = mi_label)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("No" = "lightsteelblue", "Yes" = "indianred")) +
  labs(title = "Pulse wave velocity by MI label (PPG feature)",
       x = "PWV (m/s)", y = "Density", fill = "MI") +
  theme_classic()
```

Demographics separate the classes weakly. Age trends higher in MI patients, as expected clinically, but the distributions overlap heavily. A demographics-only model should struggle, and the contrast between its weakness and the PPG channel's strength is what the ablation is designed to expose.

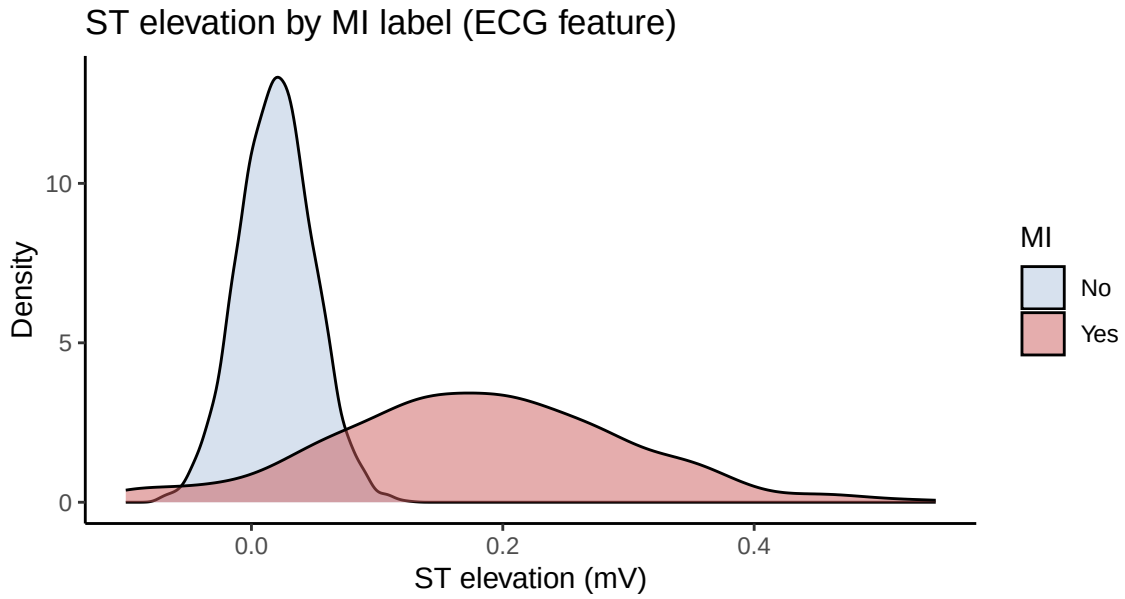


Figure 2: ST elevation by MI label. The near-separation of these distributions is why any model with ECG access classifies MI almost perfectly. This is a feature of the synthetic data, not leakage.

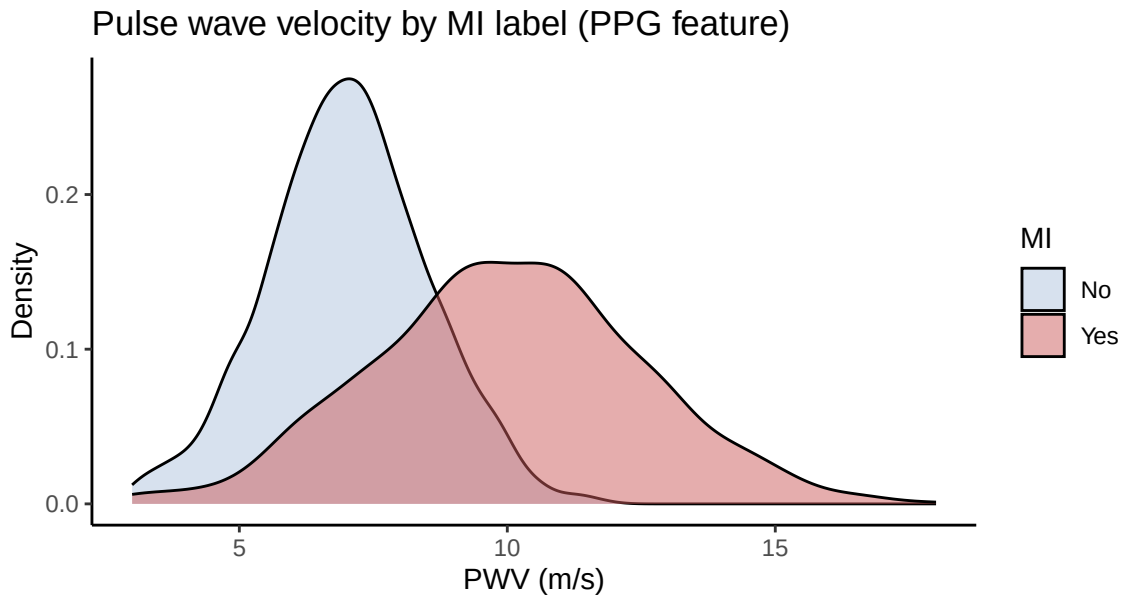


Figure 3: Pulse-wave velocity by MI label. The PPG channel separates the classes meaningfully but with more overlap than ECG, previewing strong-but-imperfect PPG-only performance.

```
ggplot(df, aes(age, fill = mi_label)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("No" = "lightsteelblue", "Yes" = "indianred")) +
  labs(title = "Age by MI label (demographic feature)",
       x = "Age", y = "Density", fill = "MI") +
  theme_classic()
```

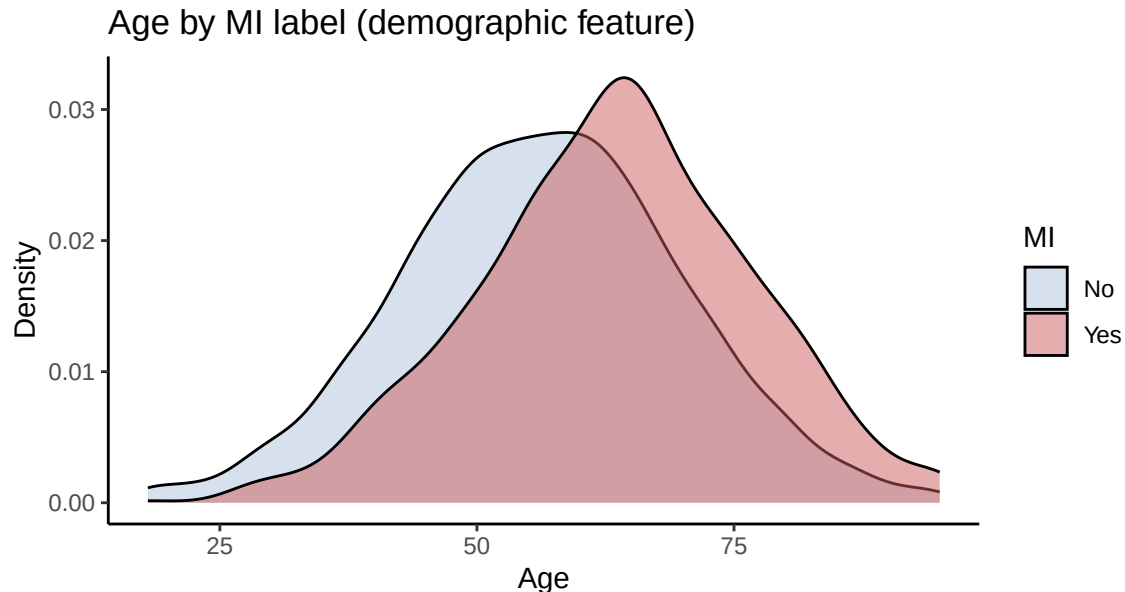


Figure 4: Age by MI label. Demographic risk factors shift in the expected direction but overlap heavily, foreshadowing weak demographics-only performance.

2.5 Train/test split

The data is split into 80% training and 20% test, stratified on the outcome so both partitions preserve the roughly 18.5% MI prevalence. The split ratio is deliberate. With 5,000 rows, a 20% test set holds about 1,000 patients and roughly 185 MI cases. That's enough that minority-class metrics like sensitivity are estimated stably and do not swing on a handful of cases. Simultaneously, 4,000 rows for training gives the random forest ample data to fit. A more aggressive split such as 50/50 would needlessly starve the models of training data and the extra test precision it buys isn't worth that cost. A very lean test set such as 90/10 would leave only about 90 MI cases and make sensitivity noisy. 80/20 balances those concerns.

Note that in `caret::createDataPartition`, the argument `p` sets the proportion of data placed in the returned index. That index becomes the test set here, so `p = 0.2` produces a 20% test set. The split seed uses `sample.kind = "Rounding"` so the partition is reproducible across R versions.

```
set.seed(1, sample.kind = "Rounding")
test_index <- createDataPartition(df$mi_label, times = 1, p = 0.2, list = FALSE)
train <- df[-test_index, ]
test <- df[test_index, ]

# Align factor levels in test to those learned on train, so prediction never
# meets an unseen category. With a stratified split on 5,000 rows this is a
# safeguard rather than an expected event. The stopifnot confirms no NA results.
```

```

for (c in chr_cols) test[[c]] <- factor(test[[c]], levels = levels(train[[c]]))
stopifnot(!any(is.na(test)))

cat("train rows:", nrow(train), " test rows:", nrow(test), "\n")

## train rows: 3999 test rows: 1001

cat("train MI prevalence:", round(mean(train$mi_label == "Yes"), 4),
    " test MI prevalence:", round(mean(test$mi_label == "Yes"), 4), "\n")

## train MI prevalence: 0.1848 test MI prevalence: 0.1848

```

2.6 Modeling approach

Two algorithms are trained on each feature set. The first is logistic regression, an interpretable linear baseline appropriate for binary classification. The second is a random forest, an ensemble of decision trees that serves as the more advanced model required by the project. It captures non-linearities and feature interactions without manual specification and stays stable on the perfectly-separable predictors that make plain logistic regression numerically unreliable here.

The two models are trained on identical data under identical random seeds for every feature set. Any difference in performance is therefore attributable to the model or the features. Logistic regression is fit through caret with five-fold cross-validation on the training set. The random forest uses its own out-of-bag estimate internally. All reported numbers come from the held-out test set, which is never used in training or model selection.

Performance is summarized with the metrics that survive class imbalance. Accuracy is reported but read against the no-information rate. Sensitivity is the headline clinical metric, because missing a real heart attack (a false negative) is far costlier than a false alarm. Specificity, balanced accuracy, Cohen's kappa, and AUC round out the picture. AUC matters most because it is threshold-independent and exposes models that look acceptable on accuracy only because they default to the majority class.

```

# Build a feature set from a chosen set of buckets.
build_feature_set <- function(bucket_names) {
  unlist(buckets[bucket_names], use.names = FALSE)
}

# Five-fold CV on the training set, with class probabilities for AUC scoring.
control <- trainControl(method = "cv", number = 5,
  classProbs = TRUE, summaryFunction = twoClassSummary)

# AUC via the Mann-Whitney U relationship (no extra package dependency).
auc_score <- function(prob, truth) {
  r <- rank(prob)
  n1 <- sum(truth == "Yes")
  n0 <- sum(truth == "No")
  (sum(r[truth == "Yes"]) - n1 * (n1 + 1) / 2) / (n1 * n0)
}

# Train both models on a feature set, evaluate on test, return one tidy row each.
run_models <- function(feature_cols, set_label) {
  tr <- train[, c(feature_cols, "mi_label")]

```



```

te <- test[, c(feature_cols, "mi_label")]

set.seed(1)
glm_fit <- train(mi_label ~ ., data = tr, method = "glm",
                 family = "binomial", trControl = control, metric = "ROC")
glm_pred <- predict(glm_fit, te)
glm_prob <- predict(glm_fit, te, type = "prob")[, "Yes"]
glm_cm <- confusionMatrix(glm_pred, te$mi_label, positive = "Yes")

set.seed(1)
rf_fit <- randomForest(mi_label ~ ., data = tr)
rf_pred <- predict(rf_fit, te)
rf_prob <- predict(rf_fit, te, type = "prob")[, "Yes"]
rf_cm <- confusionMatrix(rf_pred, te$mi_label, positive = "Yes")

grab <- function(cm, prob, model) {
  data.frame(
    feature_set = set_label,
    model       = model,
    n_features  = length(feature_cols),
    accuracy    = unname(cm$overall["Accuracy"]),
    sensitivity  = unname(cm$byClass["Sensitivity"]),
    specificity  = unname(cm$byClass["Specificity"]),
    balanced    = unname(cm$byClass["Balanced Accuracy"]),
    kappa       = unname(cm$overall["Kappa"]),
    AUC         = auc_score(prob, te$mi_label),
    row.names   = NULL
  )
}
rbind(grab(glm_cm, glm_prob, "logistic"),
      grab(rf_cm, rf_prob, "randomForest"))
}

```

On feature sets that include ECG, the classes are so nearly separable that logistic regression returns fitted probabilities of effectively 0 or 1, triggering R’s “fitted probabilities numerically 0 or 1 occurred” warning. That warning marks perfect separation rather than a failure, and it shows how cleanly the ECG features encode the outcome. The random forest does not hit this problem, which counts in its favor on this data.

2.7 The ablation

Four feature sets are compared, each defined purely by which buckets it may see:

1. Full (ECG + PPG + Demographics), everything except measurement metadata.
2. ECG only, the diagnostic channel in isolation.
3. PPG + Demographics, the central research question. This is wearable-grade measurement plus basic risk factors, with the diagnostic ECG removed.
4. Demographics only, risk factors with no recorded cardiac measurement at all, as a weak-baseline floor.

```

# No-information rate on the test set. This is the accuracy of always predicting
# the majority class, and every model's accuracy should be read against it.
NIR <- max(prop.table(table(test$mi_label)))
cat("Test-set no-information rate:", round(NIR, 4), "\n")

```

```
## Test-set no-information rate: 0.8152
```

```
model_sets <- list(
  "Full (ECG+PPG+Demo)" = c("ecg", "ppg", "demographics"),
  "ECG only"            = c("ecg"),
  "PPG + Demographics"  = c("ppg", "demographics"),
  "Demographics only"   = c("demographics")
)

results <- do.call(rbind, lapply(names(model_sets), function(label) {
  run_models(build_feature_set(model_sets[[label]]), label)
}))
```

3 Results

```
knitr::kable(results, digits = 4,
  caption = "Ablation results on the held-out test set. Accuracy is
  read against a no-information rate of 0.815, and sensitivity is
  the share of true MIs caught.")
```

Table 1: Ablation results on the held-out test set. Accuracy is read against a no-information rate of 0.815, and sensitivity is the share of true MIs caught.

feature_set	model	n_features	accuracy	sensitivity	specificity	balanced	kappa	AUC
Full (ECG+PPG+Demo)	logistic	41	0.9990	0.9946	1.0000	0.9973	0.9967	1.0000
Full (ECG+PPG+Demo)	randomForest	41	0.9980	1.0000	0.9975	0.9988	0.9934	1.0000
ECG only	logistic	21	0.9960	0.9892	0.9975	0.9934	0.9867	0.9999
ECG only	randomForest	21	0.9960	0.9946	0.9963	0.9955	0.9868	1.0000
PPG + Demographics	logistic	20	0.9750	0.9081	0.9902	0.9492	0.9155	0.9888
PPG + Demographics	randomForest	20	0.9790	0.9135	0.9939	0.9537	0.9287	0.9888
Demographics only	logistic	8	0.8182	0.0973	0.9816	0.5395	0.1157	0.7354
Demographics only	randomForest	8	0.8112	0.0541	0.9828	0.5184	0.0556	0.6969

The progression tells a clear four-tier story.

The full model and the ECG-only model are both near-perfect, with accuracy and AUC at their ceilings and sensitivity around 99%. Adding PPG and demographics on top of ECG changes almost nothing, because the ECG channel already contains nearly all the recoverable information. This confirms the framing from the overview. In this dataset, detecting MI from a clean ECG is not the challenge. Reporting that number alone wouldn't demonstrate much.

The PPG + Demographics model is the core result. With the diagnostic ECG entirely removed and even heart rate withheld, it still reaches about 97 to 98% accuracy, catches roughly 91% of true MIs, and achieves

an AUC near 0.99. The wearable-grade optical measurements, combined with basic risk factors, recover the large majority of the discriminative power of a full clinical ECG. That is the useful finding, and it suggests that in this synthetic setting, non-diagnostic sensing carries far more MI information than one might expect.

The demographics-only model exposes why accuracy cannot be trusted on its own. It posts about 81% accuracy, which sounds respectable until it is compared to the no-information rate of 0.815. The model sits at that floor. It reaches its accuracy by predicting “No” for almost everyone, catching only about 5 to 10% of actual MIs. Its kappa is near zero and its AUC, about 0.70 to 0.74, is far below the other models. This is a powerful illustration of the accuracy paradox under class imbalance. It is the clearest justification for reporting sensitivity and AUC throughout rather than accuracy alone.

Comparing the two algorithms, the random forest and logistic regression perform almost identically on the strong feature sets. The random forest is slightly ahead on sensitivity and balanced accuracy on the harder PPG + Demographics set, and it avoids the numerical-separation warnings that affect logistic regression on the ECG sets.

4 Conclusion

This project started from the HEART score and the idea of grouping cardiac predictors into clinical buckets. A first model built on the HEART-aligned features predicted MI almost perfectly, around 99.9% accuracy, which prompted the real investigation. The question was how much of that accuracy survives once the diagnostic ECG channel is removed. To answer it, I re-bucketed the predictors and ran an ablation, training logistic regression and a random forest on feature sets defined by clinical buckets and comparing them on metrics chosen to stay honest under class imbalance.

The central finding is that the photoplethysmogram channel, the kind of measurement a consumer wearable can capture, combined with basic demographics recovers most of the MI predictive power (around 91% sensitivity and AUC near 0.99) even with the diagnostic ECG and heart rate deliberately withheld. The demographics-only model, by contrast, performs no better than guessing the majority class, which doubles as a concrete demonstration of why accuracy is the wrong headline metric here.

Taken at face value, the potential impact points toward the promise of low-cost wearable sensing for cardiac risk screening. That reading must be heavily qualified, which leads to the limitations. Most important, the data is synthetic. A generator that writes the textbook ECG criteria for MI directly into the ST/T/Q-wave columns produces cleaner separation than real physiology would, so the near-perfect ECG performance and the strong PPG performance are both optimistic relative to clinical reality. The dataset also lacks the cardiac troponin blood test, along with the structured clinical context that scoring systems like the HEART score combine. Finally, the ablation treats buckets as clean modalities, but real ECG and PPG measurements share underlying physiology, so the separation between channels is cleaner in this design than it would be in practice.

The natural future work is to repeat this ablation on a real, non-synthetic cohort with troponin and outcome follow-up, to add probability-threshold tuning and cost-sensitive learning so the sensitivity and specificity trade-off can be set to a clinically appropriate operating point, and to test additional algorithms such as gradient boosting. The bucketing framework built here, defining models by transparent feature groups and comparing them on imbalance-aware metrics, transfers directly to that more demanding setting.

5 References

- Suraj, M. K. *ECG + PPG Features for MI Classification* (synthetic dataset, CC BY-SA 4.0). Kaggle.
- Irizarry, R. A. *Introduction to Data Science: Data Analysis and Prediction Algorithms with R*. <https://rafalab.dfci.harvard.edu/dsbook/>